R. Brown

Nint

~~Eight~~h Annual In-House Convention 37

18 & 19 April 1983

Room 4258/63

Department of Psychology

Dalhousie University

Ninth ~~Eighth~~ Annual In-House Convention

Department of Psychology

Dalhousie University

18 & 19 April 1983

Schedule

MONDAY, APRIL 18

1:45 - 3:15 Session I. Chair: B. Rusak ~~R. Rodger~~

R. Rodger - Welcome
1. R. Klein
2. K. Grasse
3. B. Fantie/R. Brown/W. Moger
4. J. Kruse
5. S. Nakajima

COFFEE BREAK

3:45 - 5:00 Session II. Chair: B. Matheson

6. J. Fentress
7. S. Bryson
8. E. Hansen
9. K. Murphy/D. Mitchell
10. S. Shaw

BEER IN LOUNGE

TUESDAY, APRIL 19

9:15 - 10:30 Session III. Chair: J. Marsh

11. J. Matsubara
12. L. Smith/R. Klein
13. W. Honig/W. Matheson/P. Dodd
14. B. Robertson/J. McNulty
15. A. Frohlich

COFFEE BREAK

11:00 - 12:15 Session IV. Chair. J. Barresi

16. I. Meinertzhagen
17. K. Briand
18. M. Harrington
19. R. Kruk
~~20. J. Mendelson~~

12:15 - 2:30 **LUNCH & POSTER SESSION**

Posters

20. J. Mendelson
21. G. Eskes
22. B. Fantie
23. B. Rusak
24. D. Henken/M. Yoon
25. J. Radel/M. Yoon
26. N. Swindale
27. M. Yoon/F. Baker

2:30 - 3:45 Session V. Chair: L. White

28. J. Fairless
29. J. Werker
30. D. Clarke
31. J. Ryon/P. McLeod
32. C. Shaw

3:45 - 4:15 **COFFEE BREAK**

4:15 - 5:30 Session VI. Chair: ~~B. Rusak~~ R. S. Rodger

33. A. Delamater
34. M. Cynader
35. C. Reed-Elder
36. R. Lydon/S. Nakajima
37. R. Brown / S. Douglas 5:15

5:30 **BEER & DINNER**

ABSTRACTS

1. R. Klein

Nostrils, Hemispheres and Human Performance (humans)

It has recently come to our attention that the distribution of EEG activity over the two hemispheres is tightly linked to the pattern of air flow through the two nostrils (Werntz et al., in press a,b). This was true both when air flow was allowed to vary with the nasal cycle and when subjects were instructed to manually block one nostril and breathe attentively through the other. In our investigation we will seek to determine if human performance on hemisphere specific (verbal and spatial) tasks will be appropriately influenced by which nostril is breathing.

2:15 2. K. Grasse

Cortical Input to the Accessory Optic System (cats)

2:30 3. B.D. Fantie, R.E. Brown and W. Moger

The Effects of Constant Light and Constant Dark on Sexual Behaviour and Hormone Levels in Adult Male Rats (rats)

Adult male Long-Evans rats were purchased from Canadian Breeding Farms at 90 days of age and housed under a 12:12 L:D cycle (LD, n= 11); constant light (LL, n = 13) or constant dark (red light) (DD, n = 11) for 70-90 days. More of the rats housed in LL had disrupted activity rhythms than those housed in DD. Each male was given two 30 minute sexual behaviour tests, one in their active phase and one in their inactive phase, separated by at least one week. Males in LD achieved significantly more ejaculations than males in the LL or DD. One week after the second sexual behaviour test, each male had a blood sample taken by cardiac puncture under halathane anaesthesia. Males in DD had significantly higher testosterone levels than those in LD or LL. Males housed in LL had significantly higher prolactin levels than those in LD or DD. The males were then sacrificed and the weights of their accessory sex organs taken. There were no differences among groups in weights of any organs: left and right testes; left and right epididymes; seminal vesicles or ventral prostate. These results indicte that both LL and DD disrupt sexual behaviour in male rats and may do so through different neuroendocrine mechanisms.

2:45 4. J. Kruse

Response Instructions and Choosing in Pigeons (pigeons)

Pigeons choose the correct one (e.g. sometimes red, other times green) of two colored discs on the wall of a Skinner box. Is the task easier when the pigeon must peck rapidly to red and slowly to green, than if they respond in the same way to both colors? Honig (1978) suggested that pigeons solve problems by issuing an "instruction"; remembering over a delay interval may take the form of "what I should do next", rather than what the most recent "external instruction", or so was. In this experiment, the "what next" alternatives are less confusable with each other for one group than the other.

5. S. Nakajima

Effect of Metergoline on Self-Stimulation

Rats responding for habenular stimulation completely stopped responding after injection of metergoline, a serotonergic blocking agent. The drug has very little effect on the same animal making the same response for hypothalamic stimulation. In contrast, chlorpromazine, which interferes with catecholamine transmission, reduced responding for hypothalamic stimulation and responding for habenular stimulation almost equally. The results suggest that serotonergic neurons are involved in the generation of rewarding effect from the habenula.

6. J.C. Fentress

345 In (Limited) Praise of Subjectivity

Animals are furry, scaly, sometimes slimy projective tests. It is notoriously easy to impute to them mental states that belong more properly to us. On the other hand, skilled (practiced) observers can often get an accurate "sense" of what animals will do in a variety of contexts, without being able to articulate the objective bases for any given prediction. One may thus code behaviour at various levels, only some of which follow traditionally accepted cognitive constraints. By attempting to make our "senses sensible" (objective) we might broaden our analytical framework. Wolves - and friends - provide my own projective test here.

400 7. S. Bryson

The Development of Writing Posture in Left-Handed Children and its Relation to Reading Skills

This report provides data on the development of writing posture in left-handed children. It extends earlier work by including careful measures of children who deviate from the standard inverted and noninverted postures. This involved the development of a scale for rating degree of inversion/noninversion based on position of the hand and pencil. We also assessed the relationship between left-handed writing postures and reading skills. Children who persisted in using a parallel posture had lower reading scores than both their inverted and noninverted peers. The results are discussed in terms of Levy's (1976, 1980) idea that handedness and writing posture are an index of the lateralization of cognitive functions.

8. E. Hansen

Research Issues in Shifts of Visual Attention

This presentation will be about shifting visual attention, that is, the focusing of attention away from fixation. Some theoretical and practical problems in this area will be reviewed. Variations in simple experimental parameters (e.g. stimulus-response probability, shift-cue location) have impact on performance measures of shift effects. The extent of this impact will also be discussed. On-going research deals with the effects of stimulus and response attribute differences on shift effects, and examines these through a choice response time paradigm. A brief description of this research will be presented.

9. K. Murphy and D.E. Mitchell

The Labile Nature of the Visual Recovery Promoted by Reverse Occlusion in Monocularly Deprived Kittens (cats)

Although the behavioural and physiological effects of early monocular deprivation can be extremely severe, they are not necessarily irreversible. Substantial recovery can occur upon restoration of visual input to the deprived eye, particularly if the nondeprived eye is occluded at the same time (reverse occlusion). Because of its relevance to conventional patching therapy of amblyopia, we have examined the permanence of the behavioural recovery promoted by reverse occlusion of monocularly deprived kittens. Surprisingly, upon termination of reverse occlusion (and restoration of visual input to both eyes) there was a rapid and reciprocal change in the visual acuity of the two eyes: much of the gains in the vision of the initially deprived eye were lost within 10 days of termination of reverse occlusion. This suggests a fundamental difference between the nature of the central connections established during the initial period of monocular occlusion and those formed during the period of reverse occlusion.

10. S. Shaw

Pupil Responses in Pupil-Less Eyes (flies)

The pupil response (PR) in insect eyes is a rapid, profound darkening brought about by illumination, supposedly caused by migration of screening pigment granules (SPGs) inside the photoreceptors. Incidental observations revealed surprisingly large changes in optical transmission (>90%), in eye mutants largely lacking pigments (<1% normal). In EM sections, SPGs are largely absent, suggesting that the conventional explanation for the PR is incorrect. Suggestively, mitochondria congregate distally in certain photoreceptors, with a distribution characteristic of the PR. The pupil pigment, possibly associated with these mitochondria, dissolves away during EM fixation.

11. J. Matsubara

Clustering of Local Connections Within the Visual Cortex (cats)

12. L. Smith and R. Klein

Semantic Priming (humans)

The time-course of accessing information from semantic memory was investigated by preceding a target stimulus for a lexical decision (word-nonword) with a prime word over a range (-40 to +640 msec) of temporal intervals. With respect to target words, the prime word was either semantically related, unrelated or neutral. Three sets of materials were used to vary the type of semantic relation (i.e., categorical, associative, or indirect) between the prime and target. Generally, a related prime facilitated decision times to words, and an unrelated prime inhibited decision times, relative to the neutral condition. The magnitude and time-course of these effects were different for three sets of materials. We take all of these effects to have implications for the nature of the retrieval mechanisms and structure of semantic memory.

13. W.K. Honig, W. Matheson and P.W.D. Dodd

Transfer of Discrimination Based on Outcome Expectancies (pigeons)

Pigeons were trained to expect food as a trial outcome following red and blue initial stimuli, and water as a trial outcome following green and white. They first learned a conditional discrimination based on red and green. A second discrimination was then introduced with blue and white. For half the birds, the correct responses in this discrimination were congruent with the outcome expectancies based on red and green. For the other half, the correct responses were incongruent. We obtained immediate transfer with the blue-white problem with congruent expectancies, and no transfer with incongruent expectancies. In fact, the introduction of the "incongruent" procedure disrupted performance of the first-learned discrimination.

14. B. Robertson and J.A. McNulty

Visual Form Perception in Hearing-Impaired Children: Perceptual Deficit or Developmental Lag? (humans)

There is evidence to suggest that differences exist between hearing-impaired and normally-hearing children in the perception of visual patterns, particularly on tasks requiring discrete elements to be organized into complete forms. Hearing-impaired adults and children have been compared to normally-hearing subjects on a variety of visual perception tasks, and the hearing-impaired show a hemispheric lateralization preference that differs from the controls': a right hemisphere preference in language processing, and a bilateralized processing of spatial or nonverbal material. Hearing-impaired children also show a superiority in colour discrimination when compared with normally-hearing children, matched for age. The main research question is to determine if these differences are due to a neurological deficit or a developmental lag. The lag may be a result of delayed lateralization and/or a lack of language experience. The present study uses a cross-sectional design, and nonverbal, coloured materials to investigate the visual perception of patterns in hearing-impaired children, compared with matched, normally-hearing controls.

15. A. Frohlich

Regulation of Synapse Numbers in Fly Photoreceptors: The Effects of Hypoinnervation (flies)

In the optic neuropile, the lamina, and the visual receptors synapse upon the first interneurons. The neurons are grouped into modules (cartridges) such that normally 6 receptors synapse upon 2 monopolar cells and - processes of amarcrine cells. At the edge of the lamina this pattern is less regular: while the postsynaptic number of cells stays the same the number of presynaptic visual receptors is reduced. The effects of this hypoinnervation upon number of receptor synapses, etc. are being studied.

16. I.A. Meinertzhagen

The Effects of Light- and Dark-Rearing Upon Synaptogenesis in Fly Photoreceptors (fly)

Bees reared in no light demonstrate reduced sensitivity of spectral phototactic behaviour to wavelengths in the green and a reduced lamina component of the ERG; they also have reduced populations of synapses the receptor terminals of one of their pairs of identified green-sensitive photoreceptor (Hertel, 1983). In the fly, we have examined the effects of light and dark-rearing upon the adult frequencies of photoreceptor synapses. In Musca, dark-rearing causes a small but consistent increase in synapse frequency, the opposite result to that predicted from Hertel's result in the bee. We are therefore trying to investigate whether this effect is upon the rate of synapse formation or whether upon the rate of subsequent synapse loss, i.e., whether upon the early or late phases of synaptogenesis.

17. K. Briand

Feature Integration and Attention (humans)

Recent proposals by Ann. Treisman suggest that a critical role of focused attention is the "glueing together" of independently coded or separable dimensions of an object into a unified percept. When use of attention is prevented, dimensions will be combined incorrectly, leading to the perception of "illusory conjunctions"; combinations of dimensions which were not in fact present. An attention priming paradigm previously used in Ray Klein's lab was utilized to study the effects of spatial allocation of attention on the tendency for individuals to make such conjunctions. To this point, it seems that combinations of items which could give rise to illusory conjunctions show different effects of allocation of attention than do items which cannot give rise to such conjunctions.

18. M. Harrington

"Emotionality"

Early Weight Gain and Behavioral Responsivity as Predictors of Dietary Obesity in Rats (rats)

Female Sprague-Dawley rats were reared in litters of 4, 8, or 20. When animals reached adulthood, each animal's acoustic startle reflex, tail-pinch feeding responses and activity in an open field containing a palatable food were assessed. After completing these behavioral tests, rats were exposed to either palatable foods or a control diet for 59 days, following which all subjects were maintained on the control diet for 66 days. Rats reared in litters of 4 or 20 both gained less weight after exposure to palatable foods than did rats reared in litters of 8. Diet-induced weight gain correlated positively with acoustic startle reflex and with latency to eat in response to tail pinch.

19. R. Kruk

Relationships Between Performance in Visual Tests and Flying in an Aircraft Simulation (humans)

The results of laboratory visual tests and performance in flying an aircraft simulator were compared among student pilots, instructor pilots, and fighter pilots. The visual tests included velocity discrimination of an expanding flow pattern, manual training of motion in depth and in the frontal plane, and motion and contract thresholds for a moving square and a static sinewave grating. The simulator tasks included landing in poor visibililty, formation flight, and bombing a target at low level.

The results suggest that there exists considerable variability among pilots in ability to see a runway in poor visibility, achieve and maintain position in formation flight, and hit a target from a low level attack and that significant proportions of performance variability in these tasks may be related to individual variations in ability to discriminate flow pattern velocity and manually track visual motion vs. measured by laboratory tests.

posters

20. J. Mendelson

Spatial and Temporal Disparity in Cat Primary Auditory Cortex (cats)

21. G. Eskes

Neural Control of Hamster Seasonal Breeding Cycles (hamster)

The present experiment was designed to examine the role of prolactin (PRL) and neural mechanisms within the suprachiasmatic nuclei (SCN) and adjacent dorsal periventricular area (PVA) in the control of testicular regression in hamsters during short-day exposure. Sham-operated hamsters, hamsters bearing lesions of the SCN or the PVA and hamsters implanted with an anterior pituitary under the kidney capsule to provide high PRL, were exposed to either long or short days. After 9 weeks, several reproductive parameters were measured. SCN or dorsal PVA ablation eliminated short day-induced gonadal regression and the accompanying reduction in pituitabry and testis function. Hamsters with exogenous PRL exhibited substantial, but incomplete gonadal regression. These data indicate that gonadal regression is effected either by extra-SCN structures or by axons coursing dorsally from the SCN, and that changes in testicular function during short-day exposure require the inhibition of PRL.

22. B. Fantie

Converging Entorhinal and Septal Inputs to the Hippocampus: Evidence for an Associational Mechanism (rats)

The effect of simultaneous high frequency stimulation (HFS) delivered to both the perforant path (PP) and the medial septum (MS) on the field potential subsequently evoked by a test stimulus (applied to the PP) or a pair of test stimuli (applied to both MS and PP) recorded from the fascia dentata of six pentobarbital anaesthetized rats was examined. When PP and MS pulses were paired the septal modulation of the population spike without alteration of the EPSP was dramatically enhanced only in the hemisphere in which the MS and PP HFS had been synchronous.

23. B. Rusak

Neurophysiological Studies of the Ventral Lateral Geniculate-Suprachiasmatic Nucleus Projection in the Rat (rats)

Both the ventral nucleus of the lateral geniculate complex (LGN) and the suprachiasmatic nuclei (SCN) of the hypothalamus receive direct retinal projections. There is also anatomical evidence for a projection from LGN to SCN in rats. These projections appear to be important for mediating entrainment of circadian rhythms to illumination cycles. We used neurophysiological techniques to assess the photic responsiveness of rat SCN and LGN cells, and to determine whether a monosynaptic LGN -SCN projection exists. Our results show that both SCN and LGN cells can act as steady-state luminance detectors and that a monosynaptic projection connects these structures. This poster was presented at the 1982 Society for Neuroscience meeting and represents work done at the University of Leiden with the support of ZWO.

24. D. Henken and M. Yoon

An In Vitro Examination of Enhanced Mitosis in the Deep Layers of the Goldfish Optic Tectum During Reinnervation by Regenerating Optic Nerve Fibers (goldfish)

25. J. Radel and M. Yoon

Ultrastructural Studies on the Time Course of Invasion and Synaptogenesis by Regenerating Optic Fibers in the Optic Tectum of Adult Goldfish (goldfish)

26. N. Swindale

Visual Deprivation and Cortical Development (rats)

If visual experience is prevented, ocular dominance patches formed by geniculate afferents in layer IV of the visual cortex fail to develop properly. Short periods of deprivation (6 weeks) can be reversed by subsequent visual experience, but a longer period (7 months) cannot.

27. M.Yoon and F. Baker

Neurites Outgrowing From the Retinal Explant Prefer Co-Cultured Tectal Tissue To Cerebellar Tissue In Vitro (goldfish)

28. J. Fairless

Context Effects in Approach-Withdrawal Behavior in the Pigeon: Part II (pigeon)

29. J. Werker

Developmental Changes in Speech Perception (humans)

Previous work has indicated that infants discriminate speech sounds according to phonetic category without prior specific language experience, whereas adults and children are only sensitive to native-language (phonemic) distinctions. The present work validated the generality of these earlier findings, and showed that the reorganization in cross-language speech perception occurs around the end of the first year of life. Recent findings suggest that this pattern of initial phonetic organization followed by a subsequent broadening of category boundaries may be absent in certain language disabled individuals.

30. D. Clarke

High Frequency Cochleotopic Organization in Cat Auditory Cortex (cat)

315 31. J. Ryon and P. McLeod

The Control of Rabies in the Maritimes (foxes, raccoons,)

Rabies is a widespread, nearly always fatal disease of the CNS which can affect any warm blooded animal. The disease is maintained and spread by certain species of wildlife. Control strategies have increasingly taken advantage of behavioral research on these wild vectors, however as yet there are no effective methods of control. Research being conducted in North America and Europe focusing on a) vaccination, b) culling, and c) reproductive control of these vectors will be reviewed. This research and the present status of rabies in the Maritimes will be discussed.

32. C. Shaw

Neuroplasticity: Role of Pre and Postsynaptic Mechanisms (cats) dark reared cats

33. A. Delamater

The Role of Inter-Element Associations in the Development of Pavlovian Conditioned Inhibition (rats)

34. M. Cynader

Functional Maps in the Visual Cortex (cat)

It is now well known that cells in visual cortex respond to stimuli of particular orientations, to stimulation through one eye or another, and to stimuli which either approach or recede from the observer or which rotate from side to side. We hope to determine how cells with particular properties are laid out on the cortical surface and whether simple patterns of cortical activity result from patterns of visual input which are useful to the organism.

35. C. Reed-Elder

Inhibitory Stimulus-Reinforcer Interactions in the Piegon

Pigeons were trained to autoshape to two stimuli separately presented on a key. Independent groups received either a red houselight or a tone as a conditioned inhibitor. If the keylight illumination was accompanied by the houselight or tone, it was not followed by food presentations. During training, only one keylight stimulus was presented in compound with the putative inhibitory stimulus. Transfer of inhibitory control was tested by presenting the inhibitory stimulus in compound with each of the two keylight stimuli and measuring the number of keypeck responses. The light was expected to acquire more inhibitory strength than the tone based on stimulus-reinforcer interactions previously reported. The results confirmed this expectation.

36. R.G. Lydon and S. Nakajima

Effect of Scopolamine on Radial Maze Performance

In an 8-arm elevated radial maze, food pellets were presented at the end of 4 arms (always in the same position) but not in the remaining 4 arms. After injection of scoploamine, a cholinergic blocking agent, rats made more errors of entering into the previously entered arms. The drug had no effect on the errors of entering unbaited arms. The results suggest that "working memory" depends on cholinergic neurons but "reference memory" does not.

51 37. R. Brown

Effect of Pup Age on Paternal Behaviour of Male Long-Evans Rats

Adult male Long-Evans rats were presented with infants for 24 hr/day for 21 days. Each male received 4 pups and pups were rotated every 12 hr. Each male received pups from three litters, with litters changed approximately every 7 days. Eight males were tested with litters from 1-7 days of age (younger pups) and seven males were tested with litters from 14-21 days of age (older pups). Males were observed for two 15 min periods each day and behaviours were recorded every 15 sec. using predominant activity sampling. Males tested with older pups spent more time touching, sniffing and licking pups than males tested with younger pups. Males with young pups showed more nest building, crouching over and carrying pups. These results indicate that the pup's age regulates, to some extent, the male's parental behaviour. Males also altered their behaviours when the litters of pups were changed, suggesting that the new litter was recognized.